

Robust Optimization for Network Routing

A Framework for Deciding Whether, When, and How to Reconfigure the Network During Traffic Evolution

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Abstract

Efficient optimization of network configurations remains a critical challenge in Traffic Engineering (TE), particularly when balancing solution quality against computational cost. Building upon recent advancements in Software-Defined Networks (SDN), this study addresses the fundamental problem of determining the optimal timing and strategy for network reconfiguration. We focus on the fundamental problem of deciding whether, when and how to reconfigure the network during traffic evolution, and we investigate two distinct approaches: a heuristic 'Cluster-First-Route-Second' method and an exact Benders decomposition framework. By comparing these methods across realistic network scenarios, we evaluate their trade-offs in terms of routing optimality, execution time, and adaptability to evolving traffic patterns. Our results indicate that the heuristic provides high-quality routing solutions across networks of varying sizes, while significantly reducing the need for frequent routing reconfigurations.

Keywords: Cluster-First-Route-Second (CFRS), Benders Decomposition, MILP, MINLP, Traffic Engineering, Software-Defined Networks (SDN), Robust Routing

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1 Introduction

Traffic Engineering (TE) is a key component for network service providers, as it enables the efficient use of network resources, optimizes performance, and reduces operational costs. The challenge arises from the highly dynamic nature of traffic, caused by routine daily fluctuations and unforeseen events, which forces a trade-off between achieving optimal routing configurations and limiting the frequency of network reconfigurations.

Software-Defined Networks (SDNs) introduce the flexibility required to overcome these limitations by allowing more frequent updates of TE policies. SDN controllers, with a global view of the network state, can run TE algorithms that continuously adapt routing decisions to the current traffic. This capability enables a more dynamic and responsive network management compared to traditional static routing approaches, where configurations remain fixed for long periods.

TE strategies can generally be grouped into three categories: dynamic, static, and semi-static approaches. Dynamic TE leverages historical traffic matrices to predict future demands and compute optimal routing configurations. However, prediction errors can lead to congestion or suboptimal performance, and frequent network reconfigurations may overload the control plane. Static TE, including oblivious or robust routing, computes a single configuration based on long-term traffic patterns, which stabilizes the network but sacrifices optimality for most traffic conditions. Semi-static TE seeks a compromise by clustering traffic matrices and assigning routing configurations per cluster. Yet, arbitrary time or traffic-based clustering can cause unnecessary reconfigurations or performance degradation.

Fig. (1) presents an example with a traffic matrix (TM) containing only two demands, highlighting the fundamental distinction between the two extremes of TE: dynamic TE, which adapts the network configuration whenever traffic varies, and static TE, which keeps the configuration unchanged regardless of traffic fluctuations.

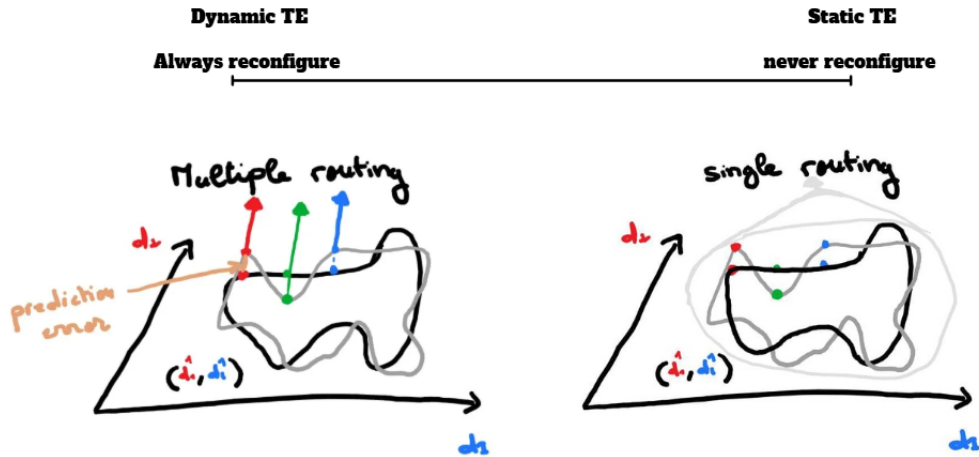


Figure 1: Dynamic TE vs. Static TE

A critical, yet frequently under-addressed aspect of SDN management involves the controller’s decision logic regarding the timing and necessity of network reconfiguration. While frequent updates maximize traffic efficiency, they risk system instability; conversely, static policies ensure stability but lead to resource underutilization. This work focuses on resolving this trade-off by identifying robust routing configurations that remain valid over clustered time intervals. We address the fundamental challenge faced by SDN controllers: determining **whether, when, and how** to reconfigure the

network following traffic changes.

To tackle this problem, we focus on constructing a set of robust routing configurations associated with clusters of traffic matrices (TMs) that overlap in time, traffic, and routing domains. Time overlap refers to the duration during which a routing configuration can be applied to TMs immediately preceding or following its cluster with minimal efficiency loss.

We revisit the Clustered Robust Routing (CRR) algorithm proposed in [Sanvito et al. \[2019\]](#), examine their heuristic in detail, and propose an exact approach based on Benders decomposition ([Benders \[1962\]](#)) after linearizing the problem formulation.

2 Summary of Contributions and Report Outline

The main contributions of this report are as follows:

- We provide new reformulations of the problem studied in [Sanvito et al. \[2019\]](#) paper, revisit the core principles of the CRR algorithm in both its overlapping and non-overlapping variants, and present a full demonstration of the objective function in the overlapping case.
- We propose an alternative solution approach, replacing the heuristic method with an exact formulation based on Benders decomposition.
- We linearize the nonlinear formulation and apply a basic Benders decomposition to the resulting linear model.
- We implement all algorithms used in our analysis, including the CRR algorithm from [Sanvito et al. \[2019\]](#), for which no code was made available. This ensures a fair comparison between the original heuristic and our exact formulations.
- We discuss the advantages and limitations of our exact approach compared with the heuristic algorithm.

The remainder of this report is organized as follows.

- Subsection [3.1](#) introduces the system model and the assumptions underlying our problem
- Subsection [3.3](#) presents heuristic approaches for constructing clustered robust routing configurations.
- Subsection [3.4](#) presents the exact formulation of the problem using Benders decomposition . We first approached the problem as a MINLP and considered generalized Benders decomposition ([Geoffrion \[1972\]](#)). Later, we decided to linearize the model, enabling a MILP formulation solvable via standard Benders decomposition ([Benders \[1962\]](#)) , which simplifies the computation without losing optimality.
- Numerical results are reported in Section [4](#).
- Section [5](#) concludes the report.
- The supplementary material section [6](#) includes a link to our codes.

3 Theoretical Framework

Tables [1–2](#) summarize the variables and notation used throughout the report.

3.1 System Model

We consider a network modeled as a directed graph $G = (\mathcal{N}, \mathcal{L})$, where $\mathcal{N}$ is the set of nodes and $\mathcal{L}$ is the set of links $e = (i, j)$, each with a capacity c_{ij} . The traffic to be routed is represented by a **Traffic Matrix (TM)**: $T(\tau) = [t_{ij}^\tau] \in \mathbb{R}^{|\mathcal{N}| \times |\mathcal{N}|}$, where t_{ij}^τ denotes the traffic from node i to node j at time τ . For simplicity, the TM can be vectorized: $D(\tau) = [d_h^\tau]$, $h \in \mathcal{H}$ where each element d_h^τ corresponds to an entry t_{ij}^τ .

The objective is to **minimize the time-averaged Maximum Link Utilization (MLU)**, defined as

$$\text{MLU}_\tau = \max_{(i,j) \in \mathcal{L}} \sum_{h \in \mathcal{H}} \frac{d_h^\tau f_{i,j}^h}{c_{i,j}}, \quad (1)$$

where $0 \leq f_{i,j}^h \leq 1$ denotes the proportion of demand d_h^τ routed through link (i, j) .

Input Parameters and Variables

Parameter	Description
$\mathcal{N}$	Set of nodes (devices)
$\mathcal{L}$	Set of links (edges)
$\mathcal{H}$	Set of demands (Origin-Destination (OD) flows)
$\mathcal{T}$	Set of time instants
$\mathcal{R}$	Set of routing configurations
c_{ij}	Capacity of link $(i, j) \in \mathcal{L}$
d_h^τ	Demand $h \in \mathcal{H}$ at time $\tau \in \mathcal{T}$
N	Number of robust routing configurations
M	Number of traffic matrices
L	Minimum duration of a configuration
O	Temporal overlap between adjacent clusters

Table 1: Input parameters of the CRR algorithm

Variable	Description
γ_r^τ	MLU when the r -th configuration is applied to $T(\tau)$
$f_{ij,h}^r$	Flow of demand h on link (i, j) under the r -th configuration
x_r^τ	Assignment of $T(\tau)$ to the r -th configuration
y_r^τ	Forward difference of x_r^τ
w_r^τ	Backward difference of x_r^τ
$\gamma_{\tau}^{-,r}$	MLU if the configuration is applied before the cluster starts
$\gamma_{\tau}^{+,r}$	MLU if the configuration is applied after the cluster ends

Table 2: CRR model variables

3.2 CRR Model: Formulation

The Clustered Robust Routing (CRR) problem seeks an optimal assignment of M Traffic Matrices (TMs) to a set of N robust Routing Configurations (RCs). The objective is to partition the time

horizon into N contiguous clusters, each subject to a minimum duration constraint L . Within each cluster, a single robust routing configuration is applied to minimize the time-averaged Maximum Link Utilization (MLU) (1). The algorithm receives as input a set of traffic matrices (TMs) that represent the period for which robust routing configurations must be designed. These TMs can be obtained in several ways: they may come from measurements of past network conditions, the output of a TM prediction module, or even be synthetically generated. We ignored the effect of prediction errors in the description of the algorithms.

Each TM represents the expected traffic conditions at a specific point in time. Consequently, the TMs can be ordered temporally, and their IDs can serve as a time axis. The algorithm produces a set of Routing Configurations (RCs, denoted as $\mathcal{R}_i$) along with the corresponding clusters of TMs ($\mathcal{C}_i$). Each RC is applied to the network as soon as the traffic enters its associated cluster.

The CRR problem is described by the following MINLP model:

$$\text{minimize } \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \gamma_r^\tau \quad (2)$$

$$\text{s.t. } \sum_{(i,j) \in \mathcal{L}} f_{ij,h}^r - \sum_{(j,i) \in \mathcal{L}} f_{ji,h}^r = b_i = \begin{cases} 1 & \text{si } i = \text{origine}(h) \\ -1 & \text{si } i = \text{destination}(h) \\ 0 & \text{sinon} \end{cases} \quad \forall i \in \mathcal{N}, \forall h \in \mathcal{H}, \forall r \in \mathcal{R} \quad (3)$$

$$\gamma_r^\tau \geq \sum_{h \in \mathcal{H}} \frac{d_h^\tau f_{ij,h}^r}{c_{ij}} x_r^\tau \quad \forall (i,j) \in \mathcal{L}, \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} \quad (4)$$

$$y_r^\tau \geq x_r^{\tau+1 \bmod |\mathcal{T}|} - x_r^\tau \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} \quad (5)$$

$$\sum_{\tau \in \mathcal{T}} y_r^\tau \leq 1 \quad \forall r \in \mathcal{R} \quad (6)$$

$$\sum_{r \in \mathcal{R}} x_r^\tau = 1 \quad \forall \tau \in \mathcal{T} \quad (7)$$

$$\sum_{\tau \in \mathcal{T}} x_r^\tau \geq L \quad \forall r \in \mathcal{R} \quad (8)$$

$$x_r^\tau, y_r^\tau \in \{0, 1\}, \quad f_{ij,h}^r \in [0, 1], \quad \gamma_r^\tau \in [0, 1] \quad (9)$$

In this model, we introduce four types of variables:

- γ_r^τ : MLU when the r -th RC is applied to $T(\tau)$.
- $f_{ij,h}^r$: Flow of demand h on link (i,j) under the r -th RC. We consider here a more general splittable routing, as it can be easily implemented in SDN switches.
- x_r^τ : binary variable indicating whether TM $T(\tau)$ is assigned to cluster r .
- y_r^τ : identifies the start of cluster r . If $y_r^\tau = 1$, then $T(\tau + 1)$ is the first element of cluster r .

Each constraint of the model is explained below:

- **(3) Flow conservation** : Standard balance constraints at each node for all demands.
- **(4) Link utilization** : Defines γ_r^τ as the maximum load (MLU) induced by routing TM τ on RC r .

- (5) **Change detection** : y_r^τ indicates a switch over time in cluster assignment.
- (6) **Single switch** : For each RC r , at most one change in cluster assignment is permitted.
- (7) **Assignment** : Each TM belongs to exactly one cluster.
- (8) **Minimum length** : Each cluster spans at least L TMs.
- (9) **Domains** : Binary variables (x_r^τ, y_r^τ) and continuous variables $(f_{ij,h}^r, \gamma_r^\tau)$.

Extension with Cluster Overlap

To account for **temporal overlap** O between adjacent clusters, the CRR model is extended so that the O overlapping TMs can be routed using either the routing configuration (RC) of their assigned cluster or that of the neighboring cluster. The MLU cost for these overlapping TMs is computed as the average between the two configurations.

By introducing a minimum **time overlap** between clusters, represented by the intersection of two clusters in Fig.(2), we ensure smoother network reconfigurations when traffic evolves into a new cluster. This approach mitigates the impact of sharp boundaries between consecutive clusters, which would otherwise require instantaneous network reconfigurations.

$$w_r^\tau \geq x_r^{\tau-1 \bmod |\mathcal{T}|} - x_r^\tau, \quad \forall \tau \in \mathcal{T}, r \in \mathcal{R} \quad (10)$$

$$\sum_{\tau \in \mathcal{T}} w_r^\tau \leq 1, \quad \forall r \in \mathcal{R} \quad (11)$$

$$\gamma_k^{-,r} \geq \sum_{h \in \mathcal{H}} \frac{d_h^k f_{ij,h}^r}{c_{ij}} \cdot y_r^\tau, \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R}, \forall (i,j) \in \mathcal{L}, \tau - O < k \leq \tau \quad (12)$$

$$\gamma_k^{+,r} \geq \sum_{h \in \mathcal{H}} \frac{d_h^k f_{ij,h}^r}{c_{ij}} \cdot w_r^\tau, \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R}, \forall (i,j) \in \mathcal{L}, \tau \leq k < \tau + O \quad (13)$$

$$w_r^\tau \in \{0, 1\}, \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} \quad (14)$$

$$0 \leq \gamma_\tau^{-,r} \leq 1, \quad 0 \leq \gamma_\tau^{+,r} \leq 1, \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} \quad (15)$$

- w_r^τ : identifies the end of a cluster to handle overlap, similar to y_r^τ for the start. If $w_r^\tau = 1$, cluster r ends at $T(\tau - 1)$.
- $\gamma_\tau^{-,r}$ and $\gamma_\tau^{+,r}$: estimate the MLU for TMs before and after a cluster ends, ensuring the adjacent cluster's RC remains feasible.

Proposition 3.1. *In the overlapping cluster case, the objective function of the CRR problem is given by :*

$$\begin{aligned} \min \quad & \sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} x_r^\tau \gamma_r^\tau + \frac{1}{2} \sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} y_r^\tau \left(\sum_{\tau - O < k \leq \tau} \gamma_k^{-,r} - \sum_{\tau + 1 \leq k \leq \tau + O} \gamma_k^r \right) \\ & + \frac{1}{2} \sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} w_r^\tau \left(\sum_{\tau \leq k < \tau + O} \gamma_k^{+,r} - \sum_{\tau - O \leq k < \tau} \gamma_k^r \right) \end{aligned} \quad (16)$$

Proof. Without overlap, the objective is simply to minimize $\sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} x_r^\tau \gamma_r^\tau$. When cluster overlap is considered, the MLU for TMs in the overlapping region is averaged between the two adjacent clusters. The objective function then becomes:

$$\sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} x_\tau^r \gamma_\tau^r + \sum_{\tau \in \mathcal{T}} \left[\sum_{r \in \mathcal{R}} y_\tau^r \left(\sum_{\tau-O < k \leq \tau} \frac{\gamma_k^{-,r} - \gamma_k^{r-1}}{2} \right) \right] + \sum_{r \in \mathcal{R}} w_\tau^r \left(\sum_{\tau \leq k < \tau+O} \frac{\gamma_k^{+,r} - \gamma_k^{r+1}}{2} \right)$$

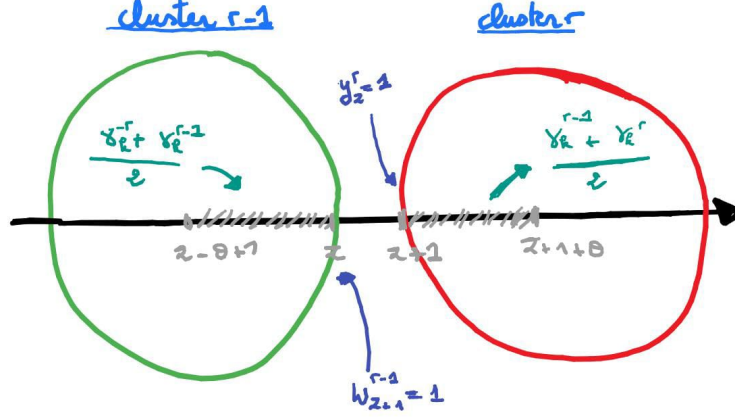


Figure 2: Illustration of overlapping clusters.

We then have:

$$\sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} y_\tau^r \left(\sum_{\tau-O < k \leq \tau} \frac{\gamma_k^{-,r} - \gamma_k^{r-1}}{2} \right) = \frac{1}{2} \sum_{\tau \in \mathcal{T}} \left(\sum_{r \in \mathcal{R}} y_\tau^r \sum_{\tau-O < k \leq \tau} \gamma_k^{-,r} - \sum_{r \in \mathcal{R}} w_\tau^r \sum_{\tau-O \leq k < \tau} \gamma_k^r \right)$$

Similarly, noting that $w_\tau^{r-1} = y_{\tau-1}^r$, we can show:

$$\sum_{r \in \mathcal{R}} w_\tau^r \left(\sum_{\tau \leq k < \tau+O} \frac{\gamma_k^{+,r} - \gamma_k^{r+1}}{2} \right) = \frac{1}{2} \left(\sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} w_\tau^r \sum_{\tau \leq k < \tau+O} \gamma_k^{+,r} - \sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} y_\tau^r \sum_{\tau+1 \leq k \leq \tau+O} \gamma_k^r \right)$$

Hence, the final objective function as presented in the paper is:

$$\begin{aligned} \sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} x_\tau^r \gamma_\tau^r + \frac{1}{2} \sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} y_\tau^r \left(\sum_{(\tau-O < k \leq \tau)} \gamma_k^{-,r} - \sum_{(\tau+1 \leq k \leq \tau+O)} \gamma_k^r \right) \\ + \frac{1}{2} \sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} w_\tau^r \left(\sum_{(\tau \leq k < \tau+O)} \gamma_k^{+,r} - \sum_{(\tau-O \leq k < \tau)} \gamma_k^r \right) \end{aligned} \quad (17)$$

This derivation ensures that each TM in the overlapping region contributes equally to both the assigned cluster and its adjacent cluster, effectively capturing the true MLU while preserving the robustness of the cluster assignments. $\square$

Remark 1. Unfortunately, the problem described by the model is highly combinatorial and has proven to be very difficult to solve. State-of-the-art integer programming solvers, such as Gurobi Solver or IBM CPLEX, could not provide a solution within reasonable time as described in (Sanvito et al. [2019]).

3.3 Heuristic Approach: Two-Step CRR Heuristic

Given the intrinsic complexity of the joint optimization task (Sanvito et al. [2019]), we adopt a heuristic framework that decomposes the global problem into more tractable subproblems. In this context, we introduce the *Clustered Robust Routing (CRR)* heuristic algorithm (Sanvito et al. [2019]), whose workflow is illustrated in Fig. 3.

The approach consists of two main phases. In the first phase, a *Segmentation Problem* is solved: starting from an initial set of W RCs, the algorithm determines an optimal assignment of M TMs to at most N of these RCs, while enforcing the minimum holding-time requirement and accounting for overlap constraints. In the second phase, a *Robust Routing Problem* is addressed for each of the N selected clusters. This step produces updated RCs that more accurately represent the behaviour of the associated TMs. These newly generated RCs are then added to the set of W available RCs used in the Segmentation Problem. The two phases are iterated for a predefined number of rounds.

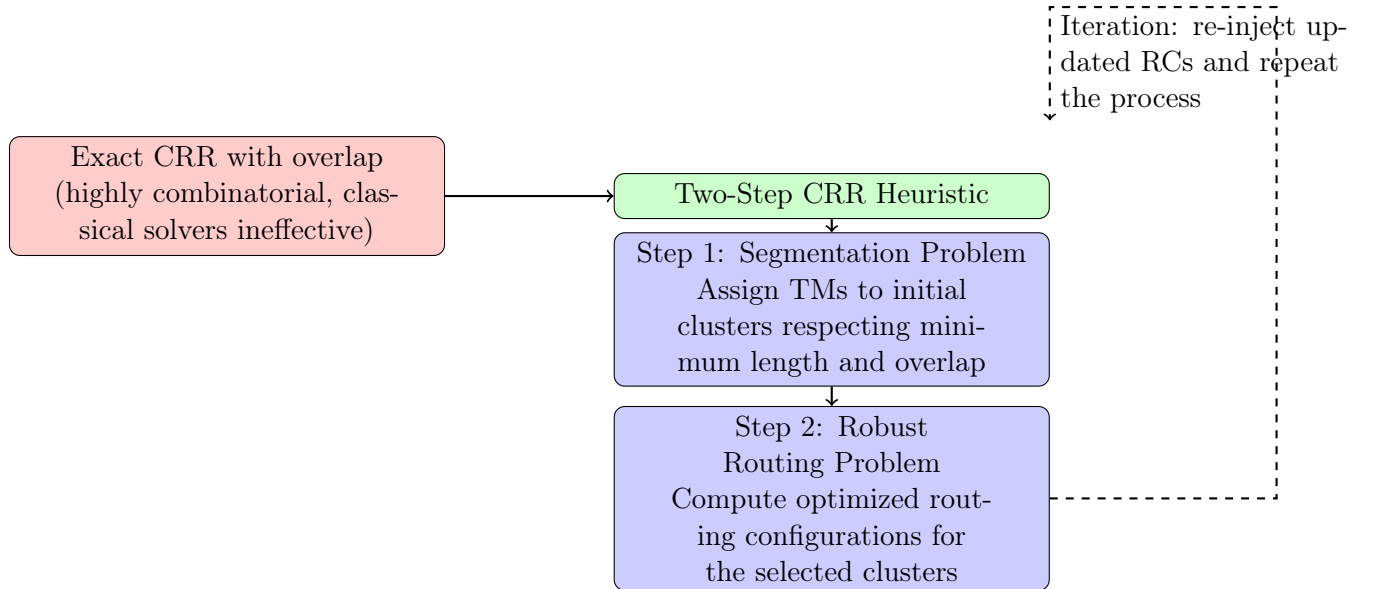


Figure 3: Workflow of the proposed two-step Clustered Robust Routing (CRR) heuristic.

3.3.1 Step 1 – Segmentation Problem (SP)

The **Segmentation Problem (SP)** takes as input a set of TMs $\mathcal{T} = \{T(1), \dots, T(M)\}$ and a set of W RCs $\mathcal{R} = \{R_1, \dots, R_W\}$. Its goal is to assign each TM $T(\tau)$ to an RC R_r so that the total assignment cost is minimized while using no more than N RCs. The TM-RC assignment cost δ_r^τ can be precomputed and corresponds to the MLU value when routing $T(\tau)$ through R_r .

Formulation without overlap

Step	Input Parameters	Decision Variables
Step 1 (SP)	• $\mathcal{T} = \{T(1), \dots, T(M)\}$: Traffic Matrices	• $x_r^\tau \in \{0, 1\}$: Assignment of TM $T(\tau)$ to RC r
	• $\mathcal{R} = \{R_1, \dots, R_W\}$: Set of candidate RCs	• $y_r^\tau \in \{0, 1\}$: Forward difference (cluster start)
	• δ_r^τ : Precomputed MLU for each TM and RC	• $w_r^\tau \in \{0, 1\}$: Backward difference (cluster end)
	• L : Minimum cluster length	• $z_r \in \{0, 1\}$: RC r selected
	• O : Overlap between clusters	
	• N : Maximum number of selectable RCs	
Step 2 (RR)	• $\mathcal{T}_c$: Set of TMs in cluster c	• $f_{ij}^h \in [0, 1]$: Fraction of demand h routed on link (i, j)
	• $\mathcal{H}$: Set of demands	
	• $\mathcal{N}$: Network nodes	• $\gamma_\tau \in [0, 1]$: Maximum Link Utilization (MLU) for TM $T(\tau)$
	• $\mathcal{L}$: Network links	
	• d_τ^h : Demand value for h at time τ	
	• c_{ij} : Link capacity of (i, j)	

Table 3: Input parameters and decision variables for the two steps of the CRR heuristic

The SP can be formulated as an ILP using a subset of the CRR variables and constraints: the variables γ_r^τ are replaced by the input parameters δ_r^τ , and unused variables ($f_{ij}^{h,r}$, γ_r^τ) are excluded. Ignoring temporarily the overlap O , the formulation is:

$$[\text{SP}] \quad \min \sum_{\tau \in \mathcal{T}, r \in \mathcal{R}} x_r^\tau \delta_r^\tau \quad (18)$$

$$\text{s.t. } y_r^\tau \geq x_r^{(\tau+1) \bmod |\mathcal{T}|} - x_r^\tau, \quad \forall \tau \in \mathcal{T}, r \in \mathcal{R} \quad (19)$$

$$\sum_{\tau \in \mathcal{T}} y_r^\tau \leq z_r, \quad \forall r \in \mathcal{R} \quad (20)$$

$$\sum_{r \in \mathcal{R}} x_r^\tau = 1, \quad \forall \tau \in \mathcal{T} \quad (21)$$

$$\sum_{\tau \in \mathcal{T}} x_r^\tau \geq L \cdot z_r, \quad \forall r \in \mathcal{R} \quad (22)$$

$$\sum_{r \in \mathcal{R}} z_r \leq N \quad (23)$$

$$x_r^\tau, y_r^\tau, z_r \in \{0, 1\}, \quad \forall \tau \in \mathcal{T}, r \in \mathcal{R} \quad (24)$$

Notes:

- $x_r^\tau = 1$ if TM $T(\tau)$ is assigned to cluster r , 0 otherwise.
- y_r^τ indicates the start of a cluster.
- z_r indicates if RC R_r is selected.
- Constraints enforce contiguous clusters of minimum length L and limit the number of selected RCs to N .

Extension for cluster overlap

To consider an overlap O between adjacent clusters:

$$w_r^\tau \geq x_r^{(\tau-1) \bmod |\mathcal{T}|} - x_r^\tau, \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} \quad (25)$$

$$\sum_{\tau \in \mathcal{T}} w_r^\tau \leq z_r, \quad \forall r \in \mathcal{R} \quad (26)$$

$$w_r^\tau \in \{0, 1\}, \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} \quad (27)$$

The objective function becomes:

$$\begin{aligned} \min \sum_{\tau, r} x_r^\tau \delta_r^\tau + \frac{1}{2} \sum_{\tau, r} y_r^\tau \left(\sum_{k=\tau-O+1}^{\tau} \delta_r^k - \sum_{k=\tau+1}^{\tau+O} \delta_r^k \right) \\ + \frac{1}{2} \sum_{\tau, r} w_r^\tau \left(\sum_{k=\tau}^{\tau+O-1} \delta_r^k - \sum_{k=\tau-O}^{\tau-1} \delta_r^k \right) \end{aligned} \quad (28)$$

Remark 2. No additional variables such as γ_r^- or γ_r^+ are needed here because δ_r^τ are precomputed constants.

3.3.2 Step 2 – Robust Routing Problem (RR)

After Step 1, TMs have been grouped into N clusters. Step 2 computes an optimized Routing Configuration (RC_{*c*}) for each cluster to better accommodate the TMs and ensure robustness against TM variations or errors.

Mathematical Formulation

For a cluster c with TMs $\mathcal{T}_c$, the robust routing problem can be formulated as:

$$[\mathbf{RR}] : \min \sum_{\tau \in \mathcal{T}_c} \gamma_\tau \quad (29)$$

where γ_τ is the Maximum Link Utilization (MLU) for TM $T(\tau)$ using RC_c .

$$\sum_{(i,j) \in \mathcal{L}} f_{ij}^h - \sum_{(j,i) \in \mathcal{L}} f_{ji}^h = \begin{cases} 1, & i = \text{origin}(h) \\ -1, & i = \text{destination}(h) \\ 0, & \text{otherwise} \end{cases}, \quad \forall i \in \mathcal{N}, \forall h \in \mathcal{H} \quad (30)$$

$$\gamma_\tau \geq \frac{\sum_{h \in \mathcal{H}} d_\tau^h f_{ij}^h}{c_{ij}}, \quad \forall \tau \in \mathcal{T}_c, \forall (i,j) \in \mathcal{L} \quad (31)$$

$$0 \leq f_{ij}^h \leq 1, \quad \forall h \in \mathcal{H}, \forall (i,j) \in \mathcal{L} \quad (32)$$

$$0 \leq \gamma_\tau \leq 1, \quad \forall \tau \in \mathcal{T}_c \quad (33)$$

Summary

- STEP 1: Group TMs into clusters and select candidate RCs.
- STEP 2: Solve a classical robust routing problem for each cluster.
- The resulting optimized RCs are fed back into STEP 1 for the next iteration of the two-step heuristic.

In short: **STEP 1 = clustering + candidate RC selection**, **STEP 2 = fine-tuned RC optimization for each cluster**.

Remark 3. Besides the classical objective of minimizing the total cost, we also formulated the problem under alternative criteria. In particular, minimization of the maximum cost and other related robustness-oriented objectives. These additional formulations were used to assess the sensitivity of the approach and to evaluate its behavior under different optimization priorities.

3.4 Exact Formulation via Benders Decomposition

To address the computational intractability of the original MINLP formulation, we propose an exact solution method based on Benders Decomposition. We initially formulated the problem using Generalized Benders Decomposition (GBD) as described by Geoffrion [1972]. However, to leverage standard solvers more effectively, we subsequently developed a linearized reformulation amenable to classical Benders Decomposition.

Remark 4. In what follows, we will focus exclusively on the version defined without the overlap between clusters. The version defined with the overlap can be treated in exactly the same way.

3.4.1 Approach I: MINLP Model with Generalized Benders Decomposition

Initial Problem

CRR Problem in GBD Form :

Define

$$X = (x_r^\tau, y_r^\tau) \in \Omega_1, \quad \text{where } \Omega_1 = \left\{ X \in \{0, 1\}^{2 \cdot |\mathcal{R}| \cdot |\mathcal{T}|} \mid \text{constraints (5)-(8) are satisfied} \right\}.$$

For a given $X = (x, y)$, we denote

$$Y = (f_{ij,h}^r, \gamma_r^\tau) \in \Omega_2(X) \subseteq \Omega_2 = \mathbb{R}_+^{|\mathcal{L}| \cdot |\mathcal{H}| \cdot |\mathcal{R}| + |\mathcal{R}| \cdot |\mathcal{T}|},$$

Where

$$\Omega_2(X) := \left\{ Y = (f_{ij,h}^r, \gamma_r^\tau) \in \Omega_2 \mid \begin{array}{l} \text{constraint (4) is satisfied by } Y \text{ for the given } X \in \Omega_1, \\ \text{constraint (3) is satisfied by } Y \end{array} \right\}.$$

The objective function is

$$f(X, Y) = \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \gamma_r^\tau.$$

With these definitions, the original CRR problem can be written as follows:

$$\inf_{X \in \Omega_1} \inf_{Y \in \Omega_2(X)} f(X, Y) \quad (34)$$

Remark 5. We can observe that X here is a vector of complicating variables, in the sense that the problem (34) reduces to a much easier optimization in Y when X is temporarily held fixed (a linear problem with no combinatorial structure).

We can reformulate the CRR problem in a Generalized Benders Decomposition (GBD) form, as introduced by Geoffrion [1972]. Constraints (4) and (3) can be written as a single vector-valued function

$$G(X, Y) \leq 0,$$

and the CRR problem in GBD form becomes

$$\begin{aligned} \min_{X \in \Omega_1, Y \in \Omega_2} \quad & f(X, Y) \\ \text{s.t.} \quad & G(X, Y) \leq 0 \end{aligned} \quad (35)$$

We can view the problem (35) using the concept of projection (partitioning) as:

$$\text{minimize } v(X) \text{ subject to } X \in \Omega_1 \cap \tilde{\Omega} \quad (36)$$

where

$$v(X) = \inf_{Y \in \Omega_2} f(X, Y) \quad \text{subject to } G(X, Y) \leq 0,$$

and

$$\tilde{\Omega} = \{X \in \Omega_1 \mid \exists Y \in \Omega_2, G(X, Y) \leq 0\}.$$

We denote, for a fixed $X \in \Omega_1$ the following problem by $A(X)$:

$$A(X) : \min_{Y \in \Omega_2} f(X, Y) \quad \text{subject to } G(X, Y) \leq 0. \quad (37)$$

$\tilde{\Omega}$ is the set of $X \in \Omega_1$ such that the problem $A(X)$ is “feasible”.

An optimal solution of (36) X^* , gives an optimal solution (X^*, Y^*) of (35) with Y^* optimal for $A(X^*)$.

Projection onto the space of X

We begin by projecting problem (35) onto the space of the variables X . The projected problem becomes

$$\min_{X \in \Omega_1 \cap \tilde{\Omega}} v(X), \quad (38)$$

where $v(X)$ denotes the optimal value of the subproblem for a fixed X .

Theorem 3.1. *Problems (35) and (38) have the same status, in the following sense:*

1. *If one of the two problems is infeasible or unbounded, then so is the other.*
2. *Let (X^*, Y^*) be an optimal solution of problem (35). Then X^* is optimal for problem (38). Conversely, if X^* is optimal for problem (38), then any Y^* attaining $v(x^*)$ in problem makes (X^*, Y^*) optimal for problem (35).*

Dual representation of $v(X)$

For $X \in \Omega_1 \cap \tilde{\Omega}$, we consider the parametrized optimization problem:

$$v(X) = \inf_{Y \in \Omega_2} f(X, Y) \quad \text{s.t.} \quad G(X, Y) \geq 0. \quad (39)$$

Our objective is to express $v(X)$ in its dual form. For fixed X , problem (39) reduces to a linear program in Y , so classical linear programming duality results apply.

For a fixed X , the value function can be written as

$$v(X) = \inf_{Y \in \Omega_2(X)} \sum_{r, \tau} \gamma_r^\tau, \quad (40)$$

where the vector of decision variables is $Y = (f_{ij,h}^r, \gamma_r^\tau) \in \Omega_2$.

Primal LP (standard form)

$$\begin{aligned} \min_{f, \gamma} \quad & \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}} \gamma_r^\tau \\ \text{s.t.} \quad & \gamma_r^\tau - \sum_{h \in \mathcal{H}} \frac{d_h^\tau x_r^\tau}{c_{ij}} f_{ij,h}^r \geq 0, \quad \forall (i, j) \in \mathcal{L}, \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R}, \\ & \sum_{(i,j) \in \mathcal{L}} f_{ij,h}^r - \sum_{(j,i) \in \mathcal{L}} f_{ji,h}^r = b_{i,h}^r, \quad \forall i \in \mathcal{N}, \forall h \in \mathcal{H}, \forall r \in \mathcal{R}, \\ & f_{ij,h}^r \geq 0, \quad \gamma_r^\tau \geq 0. \end{aligned} \quad (41)$$

Here, $b_{i,h}^r = 1$ at the origin of demand h , -1 at its destination, and 0 otherwise.

Dual variables

- Associate $\alpha_{ij,\tau,r} \geq 0$ to each inequality $\gamma_r^\tau - \sum_h \frac{d_h^\tau x_r^\tau}{c_{ij}} f_{ij,h}^r \geq 0$.
- Associate $\pi_{i,h}^r \in \mathbb{R}$ (free sign) to each flow-conservation equality $\sum_{(i,j)} f_{ij,h}^r - \sum_{(j,i)} f_{ji,h}^r = b_i$.

Lagrangian

$$\begin{aligned}\mathcal{L}_{(x,y)}(f, \gamma; \alpha, \pi) = & \sum_{r, \tau} \gamma_r^\tau - \sum_{(i,j), \tau, r} \alpha_{ij, \tau, r} \left(\gamma_r^\tau - \sum_{h \in \mathcal{H}} \frac{d_h^\tau x_r^\tau}{c_{ij}} f_{ij, h}^\tau \right) \\ & - \sum_{r \in \mathcal{R}} \sum_{h \in \mathcal{H}} \sum_{i \in \mathcal{N}} \pi_{i, h}^r \left(\sum_{(i,j) \in \mathcal{L}} f_{ij, h}^r - \sum_{(j,i) \in \mathcal{L}} f_{ji, h}^r - b_{i, h}^r \right).\end{aligned}$$

Expanding and regrouping by primal variables:

$$\begin{aligned}\mathcal{L}_{(x,y)}(f, \gamma; \alpha, \pi) = & \sum_{r, \tau} \gamma_r^\tau \left(1 - \sum_{(i,j) \in \mathcal{L}} \alpha_{ij, \tau, r} \right) \\ & + \sum_{(i,j) \in \mathcal{L}} \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} f_{ij, h}^r \left(\sum_{\tau \in \mathcal{T}} \alpha_{ij, \tau, r} \frac{d_h^\tau x_r^\tau}{c_{ij}} - (\pi_{i, h}^r - \pi_{j, h}^r) \right) \\ & + \sum_{r \in \mathcal{R}} \sum_{h \in \mathcal{H}} \sum_{i \in \mathcal{N}} \pi_{i, h}^r b_{i, h}^r.\end{aligned}\tag{42}$$

Dual LP

$$\begin{aligned}\max_{\alpha, \pi} \quad & \sum_{r, h, i} b_{i, h}^r \pi_{i, h}^r \\ \text{s.t.} \quad & \sum_{(i,j) \in \mathcal{L}} \alpha_{ij, \tau, r} \leq 1, \quad \forall r \in \mathcal{R}, \forall \tau \in \mathcal{T} \\ & \pi_{i, h}^r - \pi_{j, h}^r \leq \sum_{\tau \in \mathcal{T}} \alpha_{ij, \tau, r} \frac{d_h^\tau x_r^\tau}{c_{ij}}, \quad \forall (i, j) \in \mathcal{L}, \forall h \in \mathcal{H}, \forall r \in \mathcal{R}, \\ & \alpha_{ij, \tau, r} \geq 0, \quad \pi_{i, h}^r \in \mathbb{R}.\end{aligned}\tag{43}$$

Using the origin/destination of demand h , the dual objective can be written as:

$$\sum_{r \in \mathcal{R}} \sum_{h \in \mathcal{H}} (\pi_{\text{origin}(h), h}^r - \pi_{\text{destination}(h), h}^r).\tag{44}$$

We suppose for the rest that the CRR probleme is faisable, otherwise, there is no need for decomposition (Remark: In our experiments, the CRR is always feasible with γ only non-negative, as our networks will be undirected graphs).

Theorem 3.2 (Feasibility of the primal problem). *For any fixed $X \in \Omega_1$, the primal problem is feasible.*

Proof. In fact, for each demand h and for each $r \in R$, it is straightforward to find a flow that satisfies the demand. One can always identify a simple path from $\text{origin}(h)$ to $\text{destination}(h)$, since the CRR is feasible. The flow constraints can then be satisfied by setting $f_{i,j,h} = 1$ along the arcs of the identified path. Afterwards, taking γ as the maximum of the quantities on the left-hand side yields a feasible pair for the primal LP. $\square$

Theorem 3.3 (Equality and finiteness of primal and dual values). *For any fixed $X \in \Omega_1$, the optimal values of the primal and dual problems are equal and finite:*

$$\text{val}(\text{primal}) = \text{val}(\text{dual}) < +\infty.$$

Proof. By Theorem 3.2, the primal problem is feasible for any $X \in \Omega_1$. Then, by strong duality of linear programming, the optimal value of the primal problem coincides with the optimal value of the dual problem:

$$\text{val}(\text{primal}) = \text{val}(\text{dual}).$$

Moreover, since the primal objective is bounded below by 0, it follows that the common optimal value is finite. Therefore, both the primal and dual optimal values are equal and finite. $\square$

For a given $X \in \Omega_1$, we can write

$$V(X) = \sup_{\alpha \geq 0, \pi \geq 0} \inf_{f \geq 0, \gamma \geq 0} \mathcal{L}(f, \gamma; \alpha, \pi)$$

Remark 6. *The quantity above can be interpreted as the dual representation of $v(X)$.*

By combining all the results above, we obtain a reformulation of the initial problem, which is equivalent to the original formulation, written as:

$$(P_I) \quad \inf_{x, y \in \Omega_1} \sup_{\alpha \geq 0, \pi \geq 0} \inf_{f \geq 0, \gamma \geq 0} \mathcal{L}(f, \gamma; \alpha, \pi)$$

Hence, (P_I) can be equivalently written as:

$$(\widetilde{P_I}) \quad \min_{x, y \in \Omega_1} \theta \quad \text{s.t.} \quad \theta \geq \inf_{f \geq 0, \gamma \geq 0} \mathcal{L}(f, \gamma; \alpha, \pi) \quad \forall \pi \geq 0, \forall \alpha \geq 0.$$

Since these constraints in u and v are infinite, we will only generate a finite set of values (α^k, τ^k) .

For a fixed X obtained at the previous iteration (i.e., the solution of $(P_{I_{k-1}})$), we solve the dual problem to compute the corresponding dual variables α_k and π_k .

Consequently, at iteration k , we solve the following problem:

$$(P_{I_k}) \quad \inf_{x, y \in \Omega_1} \theta \quad \text{s.t.} \quad \theta \geq \inf_{f, \gamma \geq 0} \mathcal{L}(f, \gamma; \alpha_i, \pi_i), \quad \forall i \in \{1, \dots, k\}.$$

We refer to the following problem as the *master problem at iteration k* , denoted by MP_k :

Remark 7. *The constraints added in the above problem are referred to as optimality cuts. In a standard Benders decomposition, one would also introduce feasibility cuts to ensure primal feasibility. However, in our case, this is not necessary since, by the theorem 3.3, both the primal and dual problems are always feasible.*

To conclude, the original problem can be written as:

$$\inf_{X \in \Omega_1} \underbrace{\inf_{Y \in \Omega_2(X)} \sum_{r, \tau} \gamma_r^\tau}_{\text{Subproblem (SP) for fixed } X=(x, y)} \quad (45)$$

For this we consider two problems: the **Master Problem (MP)** and the **Subproblem (SP)**.

Master Problem (MP)

$$\begin{aligned} & \min \theta \\ & \text{s.t. } x_r^\tau, y_r^\tau \in \Omega_1 \quad (\text{binary constraints}) \\ & \theta \geq \text{exact generated Benders optimal cuts} \end{aligned}$$

Subproblem (SP)

The subproblem, given fixed binary variables $(\bar{x}, \bar{y}) \in \Omega_1$, is formulated as follows:

$$\begin{aligned}
& \min \sum_{r, \tau} \gamma_r^\tau \\
& \text{s.t.} \quad \sum_{(i,j) \in \mathcal{L}} f_{ij}^{hr} - \sum_{(j,i) \in \mathcal{L}} f_{ji}^{hr} = \begin{cases} 1 & i = \text{origin}(h) \\ -1 & i = \text{destination}(h) \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in \mathcal{N}, \forall h \in \mathcal{H}, \forall r \in \mathcal{R} \\
& \quad \gamma_r^\tau \geq \sum_{h, (i,j)} d_\tau^h f_{ij}^{hr} \frac{\bar{x}_r^\tau}{c_{ij}} \quad \forall r \in \mathcal{R}, \forall \tau \in \mathcal{T}, \forall (i,j) \in \mathcal{L} \\
& \quad 0 \leq f_{ij}^{hr}, \quad 0 \leq \gamma_r^\tau \quad \forall (i,j) \in \mathcal{L}, \forall r \in \mathcal{R}, \forall \tau \in \mathcal{T}, \forall h \in \mathcal{H}
\end{aligned}$$

Theorem 3.4 (Bounds of the Benders Decomposition). *Let θ^* denote the optimal value of the Master Problem (MP), and let γ^* denote the optimal value of the Subproblem (SP) corresponding to a given feasible solution of the MP. Then:*

1. The Master Problem provides an **lower bound** on the optimal value of the original problem:

$$\theta^* \leq z^*,$$

where z^* is the optimal value of the original problem.

2. The Subproblem provides a **upper bound** on the optimal value for a fixed binary solution $(\bar{x}, \bar{y})$:

$$\gamma^* \geq z^*.$$

Proof. The proof follows directly from the properties of Benders decomposition. The Master optimizes over a larger feasible set, yielding a lower bound, while the Subproblem fixes the integer variables, leading to an upper bound on the original problem's optimal value. $\square$

The original problem is a large-scale mixed-integer program with non-linear constraints, which makes it computationally challenging to solve exactly. In particular, even with the generalized Benders decomposition, the MP problem is not trivial to solve in practice: while it may be tractable in theory, its implementation can be computationally demanding.

To illustrate this, consider the MP problem at iteration 1. For a fixed X obtained at the previous iteration (i.e., the solution of (P_{I_0})), we solve the dual problem of the SP problem to compute the corresponding dual variables α_0 and π_0 , the MP problem at iteration 1 is given by :

$$\inf_{x, y \in \Omega_1} \theta \quad \text{s.t.} \quad \theta \geq \inf_{f, \gamma \geq 0} \mathcal{L}_{(x,y)}(f, \gamma; \alpha_0, \pi_0)$$

For a fixed $(x, y) \in \Omega_1$, we have that :

$$\begin{aligned}
\inf_{f, \gamma \geq 0} \mathcal{L}_{(x,y)}(f, \gamma; \alpha_0, \pi_0) &= \sum_{r, \tau} \gamma_r^\tau \left(1 - \sum_{(i,j) \in \mathcal{L}} \alpha_{0,ij,\tau,r} \right) \\
&+ \sum_{(i,j) \in \mathcal{L}} \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} f_{ij,h}^r \left(\sum_{\tau \in \mathcal{T}} \alpha_{0,ij,\tau,r} \frac{d_h^\tau x_r^\tau}{c_{ij}} - (\pi_{0,i,h}^r - \pi_{0,j,h}^r) \right) \\
&+ \sum_{r \in \mathcal{R}} \sum_{h \in \mathcal{H}} \sum_{i \in \mathcal{N}} \pi_{0,i,h}^r b_i.
\end{aligned}$$

$$= \begin{cases} \delta_{\text{opt}}(\alpha_0, \tau_0) & \text{if } \pi_{0,i,h}^r - \pi_{0,j,h}^r \leq \sum_{\tau} \alpha_{0,ij,\tau,r} \frac{d_h^{\tau} x_r^{\tau}}{c_{ij}}, \quad \forall (i,j) \in \mathcal{L}, \forall h \in \mathcal{H}, \forall r \in \mathcal{R} \\ -\infty, & \text{otherwise.} \end{cases}$$

Where : $\delta_{\text{opt}}(\alpha_0, \tau_0) = \sum_{r \in \mathcal{R}} \sum_{h \in \mathcal{H}} (\pi_{0,\text{origin}(h),h}^r - \pi_{0,\text{destination}(h),h}^r)$.

Note that we already have that: $\forall r \in \mathcal{R}, \tau \in \mathcal{T} : \sum_{(i,j) \in \mathcal{L}} \alpha_{0,ij,\tau,r} \leq 1$, because α_0 is already a solution of the SP at iteration 0 for some fixed X in the master problem MP_0 .

This form of the Lagrangian depends on whether the dual constraints are satisfied. Checking these constraints while maintaining a linear programming formulation necessitates the introduction of additional binary variables, which renders the problem more complex and combinatorial again...

The nonlinear link capacity constraints in the original model present significant computational hurdles. To mitigate this, we introduce a linear reformulation using a constant denoted as A . We demonstrate that for a sufficiently large value of A , the linearized constraints are mathematically equivalent to the original nonlinear set, thereby allowing the use of classical Benders cuts without loss of optimality.

3.4.2 Approach II: Linear Reformulation and Classical Benders Decomposition

Given the complexity of implementing a Generalized Benders Decomposition for our original nonlinear problem, we explored an alternative approach. Specifically, we considered a linear reformulation of the problem, which allows us to apply the classical Benders Decomposition. As described in Section 3.2, the CRR problem is formulated in Equation (2).

$$\begin{aligned} & \text{minimize } \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \gamma_r^{\tau} && \text{(Objective)} \\ & \text{s.t. } \sum_{(i,j) \in \mathcal{L}} f_{ij,h}^r - \sum_{(j,i) \in \mathcal{L}} f_{ji,h}^r = b_i = \begin{cases} 1 & \text{if } i = \text{origin}(h) \\ -1 & \text{if } i = \text{destination}(h) \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in \mathcal{N}, \forall h \in \mathcal{H}, \forall r \in \mathcal{R} && \text{(Flow conservation)} \\ & \gamma_r^{\tau} \geq \sum_{h \in \mathcal{H}} \frac{d_h^{\tau} f_{ij,h}^r}{c_{ij}} x_r^{\tau} \quad \forall (i,j) \in \mathcal{L}, \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} && \text{(Link capacity)} \\ & y_r^{\tau} \geq x_r^{\tau+1} - x_r^{\tau} \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} && \text{(Cluster change)} \\ & \sum_{\tau \in \mathcal{T}} y_r^{\tau} \leq 1 \quad \forall r \in \mathcal{R} && \text{(One change per RC)} \\ & \sum_{r \in \mathcal{R}} x_r^{\tau} = 1 \quad \forall \tau \in \mathcal{T} && \text{(Assignment)} \\ & \sum_{\tau \in \mathcal{T}} x_r^{\tau} \geq L \quad \forall r \in \mathcal{R} && \text{(Minimum length)} \\ & x_r^{\tau}, y_r^{\tau} \in \{0, 1\}, \quad f_{ij,h}^r \in [0, 1], \quad \gamma_r^{\tau} \in [0, 1] && \text{(Variable domains)} \end{aligned}$$

Among the constraints, the link capacity constraint is nonlinear. To simplify the implementation and allow the use of classical Benders decomposition, we propose to linearize this constraint. Let us

consider this linearized constraint as follows:

$$\gamma_r^\tau \geq \sum_{h \in \mathcal{H}} \frac{d_h^\tau f_{ij,h}^r}{c_{ij}} - A \cdot (1 - x_r^\tau), \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R}, \forall (i, j) \in \mathcal{L} \quad (46)$$

The linearization involves a constant A , which will be carefully chosen in a later step. Under an appropriate choice of A , we will show that The original nonlinear problem under constraints (3)–(9) and the linearized problem under constraints (3), (46), and (5)–(9) are equivalent in terms of the optimal objective value. A proof of this equivalence will be provided once A is fixed.

Theorem 3.5 (Equivalence of Nonlinear and Linearized Problems). *The original nonlinear problem under constraints (3)–(9) and the linearized problem under constraints (3), (46), (7)–(9) are equivalent in terms of the optimal objective value, for a suitably chosen constant A .*

Proof. To ensure that the linearized link capacity constraint (46) correctly represents the original nonlinear constraint, we choose a sufficiently large constant A . The idea is that A must be large enough to deactivate the constraint whenever x_r^τ is set to 0.

Formally, we can select

$$A = \frac{\max_{\tau \in \mathcal{T}} \sum_{h \in \mathcal{H}} d_h^\tau}{\min_{(i,j) \in \mathcal{L}} c_{ij}}$$

Consider the two possible cases for x_r^τ :

- **Case $x_r^\tau = 0$:** The original link capacity term is

$$\sum_{h \in \mathcal{H}} \frac{d_h^\tau f_{ij,h}^r}{c_{ij}}.$$

Since $f_{ij,h}^r \in [0, 1]$, this sum is less than or equal to 1. By subtracting A in the linearized constraint

$$\gamma_r^\tau \geq \sum_{h \in \mathcal{H}} \frac{d_h^\tau f_{ij,h}^r}{c_{ij}} - A,$$

the right-hand side becomes negative. Because of the variable domain constraint $\gamma_r^\tau \geq 0$, this effectively deactivates the constraint, and it does not restrict the feasible solution.

- **Case $x_r^\tau = 1$:** Then the term $-A \cdot (1 - x_r^\tau) = 0$, and the linearized constraint reduces exactly to the original nonlinear link capacity constraint. Therefore, the feasible region remains unchanged.

□

Thus, the linearized version of our problem without overlap is given above.

$$\begin{aligned}
& \text{minimize } \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \gamma_r^\tau \\
& \text{s.t. } \sum_{(i,j) \in \mathcal{L}} f_{ij,h}^r - \sum_{(j,i) \in \mathcal{L}} f_{ji,h}^r = b_i = \begin{cases} 1 & \text{if } i = \text{origin}(h) \\ -1 & \text{if } i = \text{destination}(h) \\ 0 & \text{otherwise} \end{cases} \quad \forall i \in \mathcal{N}, \forall h \in \mathcal{H}, \forall r \in \mathcal{R} \\
& \gamma_r^\tau \geq \sum_{h \in \mathcal{H}} \frac{d_h^\tau f_{ij,h}^r}{c_{ij}} - A \cdot (1 - x_r^\tau), \quad \forall (i,j) \in \mathcal{L}, \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} \\
& y_r^\tau \geq x_r^{\tau+1} - x_r^\tau \quad \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R} \\
& \sum_{\tau \in \mathcal{T}} y_r^\tau \leq 1 \quad \forall r \in \mathcal{R} \\
& \sum_{r \in \mathcal{R}} x_r^\tau = 1 \quad \forall \tau \in \mathcal{T} \\
& \sum_{\tau \in \mathcal{T}} x_r^\tau \geq L \quad \forall r \in \mathcal{R} \\
& x_r^\tau, y_r^\tau \in \{0, 1\}, \quad f_{ij,h}^r \in [0, 1], \quad \gamma_r^\tau \in [0, 1]
\end{aligned}$$

We will reproduce the same steps as in Section 3.4.1, except that this time the Lagrangian will no longer be a conditional function of X . Note that we will not consider the constraint $\gamma \leq 1$ in the rest of this section.

We therefore obtain the following Benders decomposition:

$$\min_{x,y \in \Omega_1} \left(\min_{(f,\gamma) \in \Omega_2(x,y)} \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \gamma_r^\tau \right).$$

For fixed (x, y) , the subproblem (SP) is given by:

$$\min_{(f,\gamma) \in \Omega_2(x,y)} \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \gamma_r^\tau$$

Dual of the subproblem. The Lagrangian of the subproblem is written as

$$\begin{aligned}
\mathcal{L}_{(x,y)}(\mathbf{f}, \boldsymbol{\delta}; \mathbf{u}, \mathbf{w}, \boldsymbol{\xi}) &= \sum_{r \in \mathcal{R}} \sum_{\tau \in \mathcal{T}} \gamma_r^\tau \left(1 - \sum_{(i,j) \in \mathcal{L}} w_{\tau r}^{(i,j)} \right) \\
&+ \sum_{h \in \mathcal{H}} \sum_{(i,j) \in \mathcal{L}} \sum_{r \in \mathcal{R}} f_{ij}^{hr} \left(u_j^{hr} - u_i^{hr} + \sum_{\tau \in \mathcal{T}} \frac{d_h^\tau}{c_{ij}} w_{\tau r}^{(i,j)} + \xi_{ij}^{hr} \right) \\
&+ \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} u_{\text{origin}(h)}^{hr} - u_{\text{destination}(h)}^{hr} \\
&- \sum_{(i,j) \in \mathcal{L}} \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} \xi_{ij}^{hr} - A \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \sum_{(i,j) \in \mathcal{L}} (1 - x_r^\tau) w_{\tau r}^{(i,j)}.
\end{aligned}$$

Dual subproblem. The dual of the Benders subproblem can be written as:

$$\begin{aligned}
\max \quad & \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} \left(u_{\text{origin}(h)}^{hr} - u_{\text{destination}(h)}^{hr} \right) - \sum_{(i,j) \in \mathcal{L}} \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} \xi_{ij}^{hr} \\
& - A \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \sum_{(i,j) \in \mathcal{L}} (1 - x_r^\tau) w_{\tau r}^{(i,j)} \\
\text{s.t.} \quad & u_j^{hr} - u_i^{hr} + \sum_{\tau \in \mathcal{T}} \frac{d_h^\tau}{c_{ij}} w_{\tau r}^{(i,j)} + \xi_{ij}^{hr} \geq 0, & \forall h \in \mathcal{H}, \forall r \in \mathcal{R}, \forall (i,j) \in \mathcal{L}, \\
& \sum_{(i,j) \in \mathcal{L}} w_{\tau r}^{(i,j)} \leq 1, & \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R}, \\
& \xi_{ij}^{hr} \geq 0, & \forall (i,j) \in \mathcal{L}, \forall h \in \mathcal{H}, \forall r \in \mathcal{R}, \\
& w_{\tau r}^{(i,j)} \geq 0, & \forall (i,j) \in \mathcal{L}, \forall \tau \in \mathcal{T}, \forall r \in \mathcal{R}.
\end{aligned}$$

Master problem.

At iteration k , the master problem (MP) is formulated as:

$$\begin{aligned}
\min_{(x,y) \in \Omega_1} \quad & \theta \\
\text{s.t.} \quad & \theta \geq \mathcal{L}_{(x,y)}(f, \gamma, \bar{u}^{(i)}, \bar{w}^{(i)}, \bar{\xi}^{(i)}), \quad \forall i \in \{0, \dots, k-1\}
\end{aligned}$$

Since the optimal dual variables obtained at iteration i are, in particular, dual-feasible and satisfy all the dual constraints, the value of the Lagrangian function evaluated at these solutions is given by:

$$\begin{aligned}
\mathcal{L}_{(x,y)}(f, \gamma, \bar{u}^{(i)}, \bar{w}^{(i)}, \bar{\xi}^{(i)}) = & \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} \left(\bar{u}_{\text{origin}(h)}^{hr,(i)} - \bar{u}_{\text{destination}(h)}^{hr,(i)} \right) \\
& - A \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \sum_{(i,j) \in \mathcal{L}} (1 - x_r^\tau) \bar{w}_{\tau r}^{(i,j),(i)} \\
& - \sum_{(i,j) \in \mathcal{L}} \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} \bar{\xi}_{ij}^{hr,(i)}, \quad \forall i \in \{0, \dots, k-1\}
\end{aligned}$$

Then the master problem (MP) at iteration k , can be written as:

$$\begin{aligned}
\min_{(x,y) \in \Omega_1} \quad & \theta \\
\text{s.t.} \quad & \theta \geq \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} \left(\bar{u}_{\text{origin}(h)}^{hr,(i)} - \bar{u}_{\text{destination}(h)}^{hr,(i)} \right) \\
& - A \sum_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} \sum_{(i,j) \in \mathcal{L}} (1 - x_r^\tau) \bar{w}_{\tau r}^{(i,j),(i)} \\
& - \sum_{(i,j) \in \mathcal{L}} \sum_{h \in \mathcal{H}} \sum_{r \in \mathcal{R}} \bar{\xi}_{ij}^{hr,(i)}, \quad \forall i \in \{0, \dots, k-1\}
\end{aligned}$$

Algorithm 1 Benders decomposition for the given problem

Input: Sets $\mathcal{H}, \mathcal{R}, \mathcal{T}, \mathcal{L}$, parameters $A, f, \gamma, c_{ij}, d_h^r$, tolerance ϵ

Initialize: $k \leftarrow 0, \theta^0 \leftarrow -\infty, LB \leftarrow -\infty, UB \leftarrow +\infty$, cut set $\mathcal{C} \leftarrow \emptyset$

While $UB - LB > \epsilon$:

Solve the Master Problem (MP):

$$\min_{(x,y) \in \Omega_1} \theta \quad \text{s.t. all cuts in } \mathcal{C}$$

Let (x^k, y^k, θ^k) be the optimal solution, update $LB \leftarrow \min(LB, \theta^k)$.

Solve the dual Subproblem (SP) for (x^k, y^k) :

$$\max \sum_{h,r} (u_{\text{origin}(h)}^{hr} - u_{\text{destination}(h)}^{hr}) - \sum_{(i,j),h,r} \xi_{ij}^{hr} - A \sum_{\tau,r,(i,j)} (1 - x_r^\tau) w_{\tau r}^{(i,j)}$$

Let $(\bar{u}^{(k)}, \bar{w}^{(k)}, \bar{\xi}^{(k)})$ be the optimal dual solution Generate Benders optimality cut:

$$\theta \geq \sum_{h,r} (\bar{u}_{\text{origin}(h)}^{hr,(k)} - \bar{u}_{\text{destination}(h)}^{hr,(k)}) - A \sum_{\tau,r,(i,j)} (1 - x_r^\tau) \bar{w}_{\tau r}^{(i,j),(k)} - \sum_{(i,j),h,r} \bar{\xi}_{ij}^{hr,(k)}$$

Add this cut to the cut set $\mathcal{C}$.

Update upper bound $UB \leftarrow \min(UB, \text{optimal value of SP problem at } (x^k, y^k))$.

Update $k \leftarrow k + 1$.

Output: Optimal (x^*, y^*) with value θ^* .

4 Numerical results

4.1 Experiment framework for the heuristic algorithm

In order to evaluate the performance of the proposed optimization framework, we rely on two well-known real backbone networks from the SNDlib repository, namely the *Nobel-Germany* and *GEANT* topologies. These networks are widely used in the literature to benchmark traffic engineering and routing optimization algorithms, due to the public availability of their structural characteristics and traffic matrix datasets. The main characteristics of these two networks are given in Table 4, while their corresponding topologies are illustrated in Figures 4 and 5. For the GEANT network, due to the large number of available traffic matrices and limited computational resources, we restrict our experiments to a subset of 200 traffic matrices.

Network	Nodes	Links	Granularity	Horizon	Number of demand matrices
Nobel-Germany	17	26	1 day	28 days	28
GEANT	22	36	15 min	4 months	11460

Table 4: Characteristics of the backbone networks used in the experiments

For each network, we consider a daily operational scenario in which a sequence of TMs describes the evolution of demand throughout the day. Our methodology consists of comparing the behavior of our algorithm with optimal routing configurations obtained by optimizing each TM independently. The proposed method is implemented in Python and relies on the COIN-OR CBC solver to solve the underlying mathematical programs.

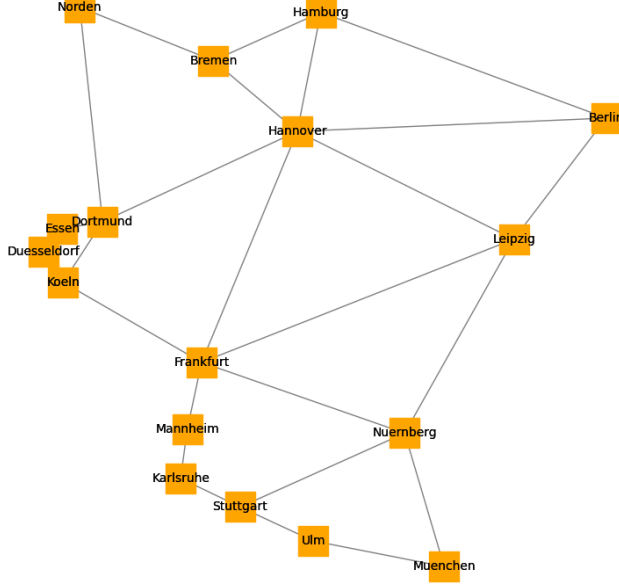


Figure 4: Topology of the Nobel-Germany network.

For a given TM at $\tau \in \mathcal{T}$, associated with a routing configuration RC r obtained through the CRR heuristic and characterized by the set of routing variables $f_{ij}^{h,r}$, we compute the *Maximum Link Utilization (MLU)* as

$$\text{MLU}_{RC}[\tau] = \max_{(i,j) \in L} \frac{\sum_{h \in H} d_k^h f_{ij}^{h,r}}{c_{ij}}. \quad (47)$$

To compare different routing policies under the same network topology and traffic conditions, we rely on the *performance ratio*, defined as

$$\text{PR} = \frac{\sum_{\tau \in T} \text{MLU}_{RC}[\tau]}{\sum_{\tau \in T} \text{MLU}_{\text{DynTE}}[\tau]} \quad (48)$$

where $\text{MLU}_{\text{DynTE}}[\tau]$ denotes the MLU obtained using Dynamic Traffic Engineering, i.e., the optimal minimum-MLU Multi-Commodity Flow (MCF) solution computed independently for each traffic matrix.

By definition, the performance ratio is always greater than or equal to 1, and measures how far a given routing strategy is from the ideal time-varying routing configuration that can adapt perfectly at each time instant.

We start with the small-scale *Nobel-Germany* network. The CRR heuristic was evaluated for different pairs (N, L) in the non-overlapping cluster setting, where N denotes the maximum number of clusters and L the minimum number of traffic matrices contained in each cluster. For each (N, L)

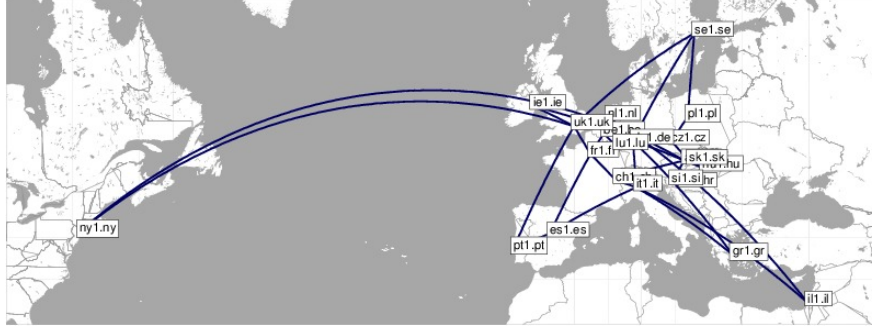


Figure 5: Topology of the GEANT network.

configuration, the heuristic was executed for 10 iterations. The corresponding results are reported in Figure 6. Overall, the performance ratio remains consistently close to 1 across all tested (N, L) configurations, indicating that the proposed heuristic achieves near-optimal solutions. Moreover, for a fixed value of L , the performance ratio decreases as N increases and stabilizes beyond a certain threshold. Conversely, for a fixed number of clusters N , the performance ratio increases with L .

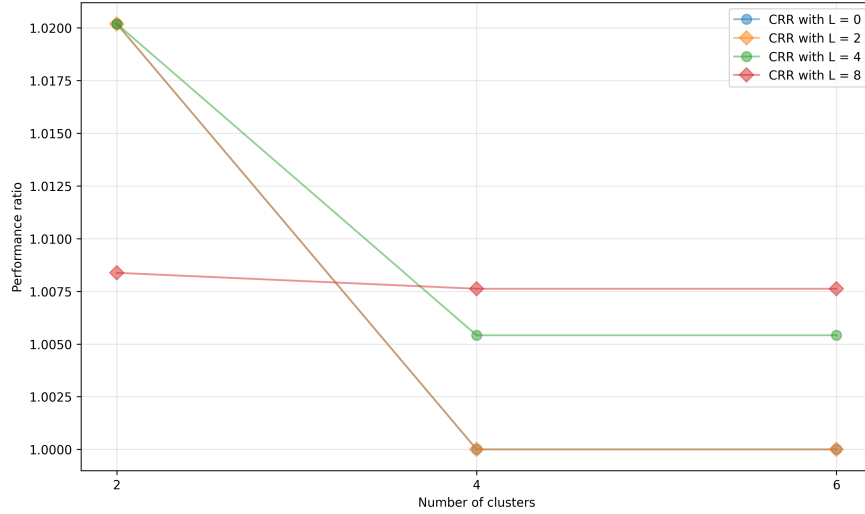
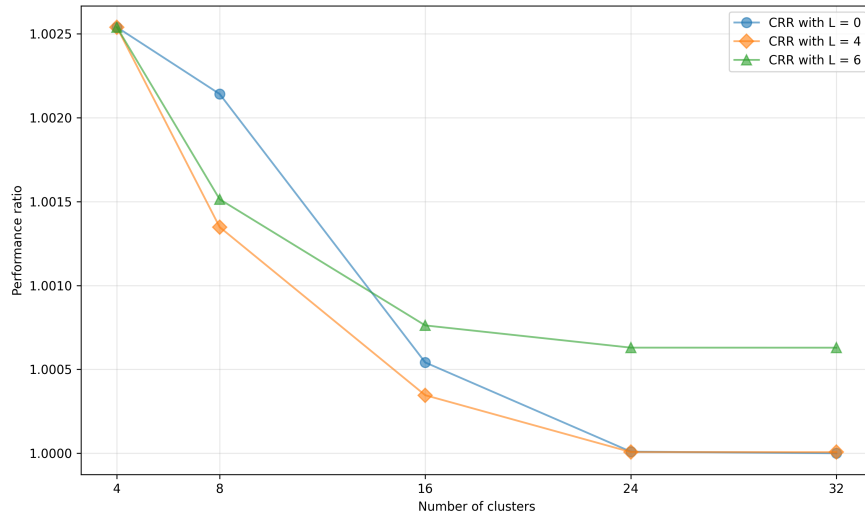


Figure 6: Performance of the CRR heuristic on the Nobel-Germany network for different (N, L) configurations (10 iterations).

To further validate these results, we now focus on the large-scale *GEANT* network. In this setting, the CRR heuristic was evaluated for different pairs (N, L) in the non-overlapping cluster setting. Due to the increased complexity of the underlying optimization models and their associated computational cost, the heuristic was executed for only 5 iterations per (N, L) configuration. The corresponding results are reported in Figure ???. As in the previous case, the performance ratio remains very close to 1 across all tested configurations, indicating that the proposed heuristic remains effective even for large-scale networks. Moreover, the performance ratio decreases with N and stabilizes beyond a certain threshold. In contrast to the *Nobel-Germany* instance, the performance ratio does not exhibit a monotonic behavior with respect to the parameter L , as can be observed from the curves corresponding to $L = 0$ and $L = 6$. Furthermore, for $L = 0$ and $L = 4$, the heuristic converges to the optimal solution (i.e., a performance ratio equal to 1) with a reduced number of clusters $N = 24$. This highlights the effectiveness of the proposed clustering approach,

as it requires updating the routing configurations associated with the traffic matrices only 24 times, while achieving the same performance as the optimal dynamic traffic engineering solution, which would require updating the routing configuration at every time step, i.e., 200 times in this case.



Performance of the CRR heuristic (5 iterations) on the GEANT network.

To further assess the effectiveness of the CRR heuristic, we also consider the *pointwise performance ratio* for each traffic matrix $\tau \in \mathcal{T}$, defined as

$$\text{PR}_\tau = \frac{\text{MLU}_{\text{RC}}[\tau]}{\text{MLU}_{\text{DynTE}}[\tau]}. \quad (49)$$

We focus on the GEANT instance with 200 traffic matrices, as considered previously, and on the specific case $L = 4$, which yields the best overall performance ratios in Figure ???. We analyze the distribution of the pointwise performance ratio across the 200 traffic matrices for different values of N , corresponding to the number of clusters. The corresponding results are reported in Figure 7. This figure further confirms the effectiveness of the proposed heuristic, as the performance ratio remains close to 1 for the majority of traffic matrices, even when using a relatively small number of clusters.

We conclude from these two experiments that the performance of the CRR heuristic consistently improves as the number of clusters N increases. In contrast, its variation with respect to the parameter L strongly depends on the specific characteristics of each network, as well as on the structural properties of their traffic matrices.

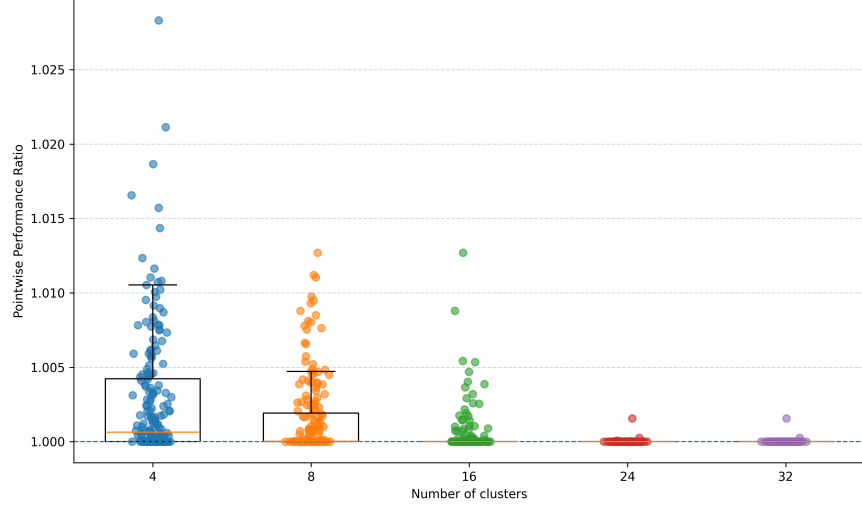


Figure 7: Distribution of the pointwise performance ratio for the GEANT network with $L = 4$, across 200 TMs and different numbers of clusters N (5 iterations).

4.2 Experiment framework for the benders decomposition algorithm

Regarding the evaluation of the performance of the Benders-based approach, we focus on the small-scale *Nobel-Germany* instance. In particular, this experiment was conducted on a reduced version of this network, considering only 5 traffic matrices, with parameters $N = 2$, $L = 2$, and under the non-overlapping cluster setting. The purpose of this reduced setting was to enable the direct computation of the optimal solution of the linearized CRR problem, so as to compare it with the solution returned by the Benders decomposition.

However, when analyzing the numerical results, we observed that the lower bound of the CRR problem didn't converge across iterations. As illustrated in Figure 8, even after 100 iterations, the lower bound does not exceed zero, which already constitutes a trivial lower bound for the CRR problem. More strikingly, the lower bound stabilizes around a negative value. In contrast, the convergence of the upper bound is significantly faster and is achieved within the first few iterations.

The lack of convergence of the lower bound can be partially explained by the introduction of the constant A in the linearized formulation, which allows the lower bound to take negative values. This observation motivates a revision of the Benders decomposition methodology adopted in this work, in order to better understand and address this behavior; this investigation is left for future work.

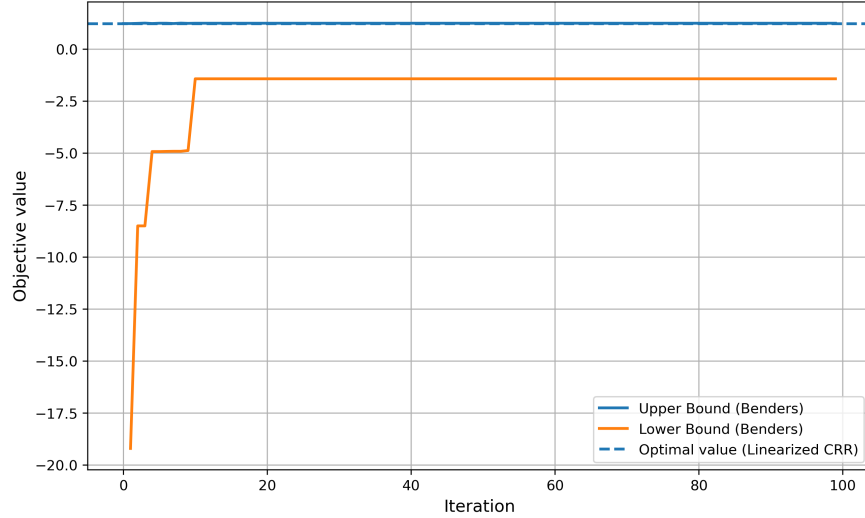


Figure 8: Evolution of the upper and lower bounds obtained by the Benders decomposition on a reduced Nobel-Germany instance, compared with the optimal value of the linearized CRR problem.

5 Conclusion

This work addressed the trade-off between routing optimality and reconfiguration frequency in Traffic Engineering for Software-Defined Networks (SDNs). Building upon the Clustered Robust Routing framework of [Sanvito et al. \[2019\]](#), we revisited its heuristic formulation and proposed an exact approach based on Benders decomposition.

Through numerical experiments on realistic backbone networks, we showed that the CRR heuristic achieves near-optimal routing performance while significantly reducing the number of routing reconfigurations compared to fully dynamic Traffic Engineering.

However, the Benders decomposition approach exhibits convergence difficulties in practice. A potential extension of this work would be to better understand the theoretical foundations of this decomposition and to identify conditions under which efficient and stable convergence can be achieved.

6 Supplementary Material

The code used to generate the experimental results presented in this report is available [here](#).

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